

Python Loops - Course Outline

Curriculum outline

Course context

Subject: Computing. Topic: Python Loops. Level: beginner. Suggested duration: 60 minutes.

The lesson focuses on for loops, while loops, range, counters. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind for loops.
- Explain the main idea behind while loops.
- Explain the main idea behind range.
- Explain the main idea behind counters.
- Apply accumulators in a short problem or example.
- Recognize a common mistake involving infinite loop debugging.

Content boundaries

Keep the lesson centered on Python Loops. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Python Loops. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: for loops, while loops, range, counters. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Python Functions - Course Outline

Curriculum outline

Course context

Subject: Computing. Topic: Python Functions. Level: beginner. Suggested duration: 60 minutes.

The lesson focuses on function definition, parameters, arguments, return values. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind function definition.
- Explain the main idea behind parameters.
- Explain the main idea behind arguments.
- Explain the main idea behind return values.
- Apply scope basics in a short problem or example.
- Recognize a common mistake involving debugging.

Content boundaries

Keep the lesson centered on Python Functions. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Python Functions. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: function definition, parameters, arguments, return values. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Python Lists - Course Outline

Curriculum outline

Course context

Subject: Computing. Topic: Python Lists. Level: beginner. Suggested duration: 60 minutes.

The lesson focuses on list creation, indexing, slicing, append/remove. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind list creation.
- Explain the main idea behind indexing.
- Explain the main idea behind slicing.
- Explain the main idea behind append/remove.
- Apply iteration in a short problem or example.
- Recognize a common mistake involving common mistakes.

Content boundaries

Keep the lesson centered on Python Lists. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Python Lists. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: list creation, indexing, slicing, append/remove. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Git Branching and Merge Conflicts - Course Outline

Curriculum outline

Course context

Subject: Computing. Topic: Git Branching and Merge Conflicts. Level: beginner. Suggested duration: 75 minutes.

The lesson focuses on branches, commits, merge, conflicts. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind branches.
- Explain the main idea behind commits.
- Explain the main idea behind merge.
- Explain the main idea behind conflicts.
- Apply pull requests in a short problem or example.
- Recognize a common mistake involving `.gitignore`.

Content boundaries

Keep the lesson centered on Git Branching and Merge Conflicts. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Git Branching and Merge Conflicts. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: branches, commits, merge, conflicts. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

SQL JOIN and GROUP BY - Course Outline

Curriculum outline

Course context

Subject: Computing. Topic: SQL JOIN and GROUP BY. Level: intermediate. Suggested duration: 75 minutes.

The lesson focuses on SELECT, JOIN, GROUP BY, COUNT. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind SELECT.
- Explain the main idea behind JOIN.
- Explain the main idea behind GROUP BY.
- Explain the main idea behind COUNT.
- Apply SUM in a short problem or example.
- Recognize a common mistake involving query debugging.

Content boundaries

Keep the lesson centered on SQL JOIN and GROUP BY. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to SQL JOIN and GROUP BY. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: SELECT, JOIN, GROUP BY, COUNT. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

REST API Basics - Course Outline

Curriculum outline

Course context

Subject: Computing. Topic: REST API Basics. Level: beginner. Suggested duration: 60 minutes.

The lesson focuses on HTTP verbs, routes, request body, response body. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind HTTP verbs.
- Explain the main idea behind routes.
- Explain the main idea behind request body.
- Explain the main idea behind response body.
- Apply status codes in a short problem or example.
- Recognize a common mistake involving validation.

Content boundaries

Keep the lesson centered on REST API Basics. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to REST API Basics. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: HTTP verbs, routes, request body, response body. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Linear Equations - Course Outline

Curriculum outline

Course context

Subject: Mathematics. Topic: Linear Equations. Level: beginner. Suggested duration: 60 minutes.

The lesson focuses on variables, inverse operations, one-step equations, two-step equations. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind variables.
- Explain the main idea behind inverse operations.
- Explain the main idea behind one-step equations.
- Explain the main idea behind two-step equations.
- Apply checking solutions in a short problem or example.
- Recognize a common mistake involving checking solutions.

Content boundaries

Keep the lesson centered on Linear Equations. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Linear Equations. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: variables, inverse operations, one-step equations, two-step equations. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Slope and Linear Graphs - Course Outline

Curriculum outline

Course context

Subject: Mathematics. Topic: Slope and Linear Graphs. Level: intermediate. Suggested duration: 75 minutes.

The lesson focuses on coordinate plane, rise/run, slope, y-intercept. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind coordinate plane.
- Explain the main idea behind rise/run.
- Explain the main idea behind slope.
- Explain the main idea behind y-intercept.
- Apply graph interpretation in a short problem or example.
- Recognize a common mistake involving graph interpretation.

Content boundaries

Keep the lesson centered on Slope and Linear Graphs. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Slope and Linear Graphs. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: coordinate plane, rise/run, slope, y-intercept. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Quadratic Expressions - Course Outline

Curriculum outline

Course context

Subject: Mathematics. Topic: Quadratic Expressions. Level: intermediate. Suggested duration: 75 minutes.

The lesson focuses on expanding, factoring, zeros, discriminant awareness. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind expanding.
- Explain the main idea behind factoring.
- Explain the main idea behind zeros.
- Explain the main idea behind discriminant awareness.
- Apply graph shape in a short problem or example.
- Recognize a common mistake involving graph shape.

Content boundaries

Keep the lesson centered on Quadratic Expressions. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Quadratic Expressions. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: expanding, factoring, zeros, discriminant awareness. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Motion, Speed, and Acceleration - Course Outline

Curriculum outline

Course context

Subject: Physics. Topic: Motion, Speed, and Acceleration. Level: beginner. Suggested duration: 75 minutes.

The lesson focuses on distance, displacement, speed, velocity. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind distance.
- Explain the main idea behind displacement.
- Explain the main idea behind speed.
- Explain the main idea behind velocity.
- Apply acceleration in a short problem or example.
- Recognize a common mistake involving graphs.

Content boundaries

Keep the lesson centered on Motion, Speed, and Acceleration. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Motion, Speed, and Acceleration. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: distance, displacement, speed, velocity. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Forces and Newton's Laws - Course Outline

Curriculum outline

Course context

Subject: Physics. Topic: Forces and Newton's Laws. Level: intermediate. Suggested duration: 75 minutes.

The lesson focuses on net force, inertia, $F=ma$, action-reaction. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind net force.
- Explain the main idea behind inertia.
- Explain the main idea behind $F=ma$.
- Explain the main idea behind action-reaction.
- Apply free-body reasoning in a short problem or example.
- Recognize a common mistake involving free-body reasoning.

Content boundaries

Keep the lesson centered on Forces and Newton's Laws. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Forces and Newton's Laws. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: net force, inertia, $F=ma$, action-reaction. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Energy and Work - Course Outline

Curriculum outline

Course context

Subject: Physics. Topic: Energy and Work. Level: intermediate. Suggested duration: 75 minutes.

The lesson focuses on work, kinetic energy, potential energy, energy transfer. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind work.
- Explain the main idea behind kinetic energy.
- Explain the main idea behind potential energy.
- Explain the main idea behind energy transfer.
- Apply simple calculations in a short problem or example.
- Recognize a common mistake involving simple calculations.

Content boundaries

Keep the lesson centered on Energy and Work. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Energy and Work. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: work, kinetic energy, potential energy, energy transfer. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Cell Structure and Function - Course Outline

Curriculum outline

Course context

Subject: Biology. Topic: Cell Structure and Function. Level: beginner. Suggested duration: 60 minutes.

The lesson focuses on cell membrane, nucleus, cytoplasm, mitochondria. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind cell membrane.
- Explain the main idea behind nucleus.
- Explain the main idea behind cytoplasm.
- Explain the main idea behind mitochondria.
- Apply plant vs animal cells in a short problem or example.
- Recognize a common mistake involving plant vs animal cells.

Content boundaries

Keep the lesson centered on Cell Structure and Function. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Cell Structure and Function. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: cell membrane, nucleus, cytoplasm, mitochondria. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

DNA, Genes, and Protein Synthesis - Course Outline

Curriculum outline

Course context

Subject: Biology. Topic: DNA, Genes, and Protein Synthesis. Level: intermediate. Suggested duration: 75 minutes.

The lesson focuses on DNA, genes, transcription, translation. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind DNA.
- Explain the main idea behind genes.
- Explain the main idea behind transcription.
- Explain the main idea behind translation.
- Apply mutation basics in a short problem or example.
- Recognize a common mistake involving mutation basics.

Content boundaries

Keep the lesson centered on DNA, Genes, and Protein Synthesis. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to DNA, Genes, and Protein Synthesis. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: DNA, genes, transcription, translation. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Photosynthesis and Respiration - Course Outline

Curriculum outline

Course context

Subject: Biology. Topic: Photosynthesis and Respiration. Level: intermediate. Suggested duration: 75 minutes.

The lesson focuses on inputs, outputs, chloroplasts, mitochondria. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind inputs.
- Explain the main idea behind outputs.
- Explain the main idea behind chloroplasts.
- Explain the main idea behind mitochondria.
- Apply energy transfer in a short problem or example.
- Recognize a common mistake involving energy transfer.

Content boundaries

Keep the lesson centered on Photosynthesis and Respiration. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Photosynthesis and Respiration. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: inputs, outputs, chloroplasts, mitochondria. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Atoms, Ions, and Isotopes - Course Outline

Curriculum outline

Course context

Subject: Chemistry. Topic: Atoms, Ions, and Isotopes. Level: beginner. Suggested duration: 60 minutes.

The lesson focuses on protons, neutrons, electrons, ions. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind protons.
- Explain the main idea behind neutrons.
- Explain the main idea behind electrons.
- Explain the main idea behind ions.
- Apply isotopes in a short problem or example.
- Recognize a common mistake involving atomic number.

Content boundaries

Keep the lesson centered on Atoms, Ions, and Isotopes. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Atoms, Ions, and Isotopes. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: protons, neutrons, electrons, ions. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Chemical Reactions and Balancing - Course Outline

Curriculum outline

Course context

Subject: Chemistry. Topic: Chemical Reactions and Balancing. Level: intermediate. Suggested duration: 75 minutes.

The lesson focuses on reactants, products, conservation of mass, coefficients. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind reactants.
- Explain the main idea behind products.
- Explain the main idea behind conservation of mass.
- Explain the main idea behind coefficients.
- Apply balancing in a short problem or example.
- Recognize a common mistake involving balancing.

Content boundaries

Keep the lesson centered on Chemical Reactions and Balancing. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Chemical Reactions and Balancing. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: reactants, products, conservation of mass, coefficients. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Acids, Bases, and pH - Course Outline

Curriculum outline

Course context

Subject: Chemistry. Topic: Acids, Bases, and pH. Level: intermediate. Suggested duration: 60 minutes.

The lesson focuses on acid/base behavior, pH scale, indicators, neutralization. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind acid/base behavior.
- Explain the main idea behind pH scale.
- Explain the main idea behind indicators.
- Explain the main idea behind neutralization.
- Apply safety in a short problem or example.
- Recognize a common mistake involving safety.

Content boundaries

Keep the lesson centered on Acids, Bases, and pH. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Acids, Bases, and pH. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: acid/base behavior, pH scale, indicators, neutralization. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Professional Email Writing - Course Outline

Curriculum outline

Course context

Subject: Business. Topic: Professional Email Writing. Level: beginner. Suggested duration: 60 minutes.

The lesson focuses on subject lines, purpose, tone, requests. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind subject lines.
- Explain the main idea behind purpose.
- Explain the main idea behind tone.
- Explain the main idea behind requests.
- Apply follow-up in a short problem or example.
- Recognize a common mistake involving clarity.

Content boundaries

Keep the lesson centered on Professional Email Writing. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Professional Email Writing. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: subject lines, purpose, tone, requests. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Giving a Short Project Update - Course Outline

Curriculum outline

Course context

Subject: Business. Topic: Giving a Short Project Update. Level: beginner. Suggested duration: 45 minutes.

The lesson focuses on context, progress, blockers, next steps. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind context.
- Explain the main idea behind progress.
- Explain the main idea behind blockers.
- Explain the main idea behind next steps.
- Apply audience in a short problem or example.
- Recognize a common mistake involving conciseness.

Content boundaries

Keep the lesson centered on Giving a Short Project Update. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Giving a Short Project Update. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: context, progress, blockers, next steps. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Meeting Notes and Action Items - Course Outline

Curriculum outline

Course context

Subject: Business. Topic: Meeting Notes and Action Items. Level: beginner. Suggested duration: 45 minutes.

The lesson focuses on decisions, owners, deadlines, open questions. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind decisions.
- Explain the main idea behind owners.
- Explain the main idea behind deadlines.
- Explain the main idea behind open questions.
- Apply follow-up in a short problem or example.
- Recognize a common mistake involving follow-up.

Content boundaries

Keep the lesson centered on Meeting Notes and Action Items. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Meeting Notes and Action Items. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: decisions, owners, deadlines, open questions. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Mean, Median, Mode, and Spread - Course Outline

Curriculum outline

Course context

Subject: Statistics. Topic: Mean, Median, Mode, and Spread. Level: beginner. Suggested duration: 60 minutes.

The lesson focuses on mean, median, mode, range. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind mean.
- Explain the main idea behind median.
- Explain the main idea behind mode.
- Explain the main idea behind range.
- Apply outliers in a short problem or example.
- Recognize a common mistake involving interpretation.

Content boundaries

Keep the lesson centered on Mean, Median, Mode, and Spread. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Mean, Median, Mode, and Spread. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: mean, median, mode, range. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Reading Charts and Tables - Course Outline

Curriculum outline

Course context

Subject: Statistics. Topic: Reading Charts and Tables. Level: beginner. Suggested duration: 60 minutes.

The lesson focuses on bar charts, line charts, percentages, comparison. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind bar charts.
- Explain the main idea behind line charts.
- Explain the main idea behind percentages.
- Explain the main idea behind comparison.
- Apply misleading axes in a short problem or example.
- Recognize a common mistake involving misleading axes.

Content boundaries

Keep the lesson centered on Reading Charts and Tables. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Reading Charts and Tables. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: bar charts, line charts, percentages, comparison. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

Water Cycle and Local Water Use - Course Outline

Curriculum outline

Course context

Subject: Environmental_Science. Topic: Water Cycle and Local Water Use. Level: beginner. Suggested duration: 60 minutes.

The lesson focuses on evaporation, condensation, precipitation, runoff. The remaining ideas are introduced after the core concepts are clear.

The lesson should move from a short concrete example into guided practice and then independent practice. Learners should explain at least one answer in their own words.

Learning objectives

- Explain the main idea behind evaporation.
- Explain the main idea behind condensation.
- Explain the main idea behind precipitation.
- Explain the main idea behind runoff.
- Apply conservation in a short problem or example.
- Recognize a common mistake involving conservation.

Content boundaries

Keep the lesson centered on Water Cycle and Local Water Use. Don't add advanced material that really belongs in a separate lesson.

New vocabulary should be defined once, then used consistently. If a formula, command, or process is introduced, include at least one worked example and one short practice item.

Include at least one misconception or near-miss that a beginner could realistically make.

Suggested lesson sequence

1. Opening

Start with a familiar situation or small problem connected to Water Cycle and Local Water Use. Ask one diagnostic question before teaching formal terms.

Keep this section short. The aim is to surface what learners already think.

2. Core explanation

Introduce the main ideas in this order: evaporation, condensation, precipitation, runoff. Use one idea at a time and keep examples small enough to trace or calculate by hand.

Avoid jumping straight to the final rule without showing why the rule is useful.

3. Guided example

Work through one complete example and explain each step. The example should be similar to, but not identical to, later practice.

Ask learners to predict the next step before revealing it where practical.

4. Practice and check

Include a mix of recognition, prediction, correction, and short construction. A learner should not be able to pass by memorizing one definition.

End with a short check that matches the learning objectives.

Assessment notes

Use 5-8 short items for a 60-75 minute lesson. Mix question types. Include at least one item where a learner must explain why an answer is correct or identify why a tempting answer is wrong.

Avoid introducing new terminology inside the assessment unless the question supplies the meaning.

01 General Lesson Design Guide

Lesson structure

A short lesson should have a clear beginning, main teaching section, guided practice, independent practice, and a final check. Not every lesson needs the same number of activities.

Examples should start simple. Add complexity after the learner has seen the core idea work at least once.

Explanations

Use plain language first, then introduce the formal term. Keep sentences short around new ideas.

Don't bury the actual rule inside a long paragraph.

Practice

A useful sequence is recognize, predict, fix, then build. This can be adapted across programming, mathematics, science, and communication skills.

Practice should resemble the lesson without copying the worked example.

Feedback

Explain the cause of an error, not only the corrected answer. When several answers are possible, describe the features of a good answer.

02 Assessment Writing Guide

Question mix

Use multiple choice, short answer, trace/predict, correction, and small creation tasks. Don't make every item a definition question.

Wrong options should represent realistic mistakes, not unrelated statements.

Difficulty

A beginner question can still require reasoning. Difficulty should come from applying what was taught, not unfamiliar vocabulary.

Intermediate questions may combine two ideas that were both covered.

Answer keys

For objective items, give the correct answer and a short reason. For open tasks, give expected behavior or marking points.

03 Beginner Vs Intermediate Guide

Beginner version

Reduce assumed prior knowledge. Use smaller examples, explicit steps, and short hints. Keep the same main idea rather than removing most of the content.

Explain what changes from one step to the next.

Intermediate version

Combine ideas learners have already seen. Ask for comparison, debugging, interpretation, or choice of method. Reduce hints without introducing unrelated advanced material.

04 Accessibility And Readability Guide

Readable material

Use descriptive headings, short paragraphs, and enough spacing between examples and explanations. Code, formulas, and key instructions should be copyable as text.

Don't rely on color alone to show correctness, categories, or sequence.

Language

Avoid unnecessary jargon. Define technical terms when they first appear. For long instructions, split the task into steps.

05 Lab And Activity Design Guide

Hands-on activities

A lab should make learners do something observable, not only read and answer. Good activities include changing one variable and comparing results, tracing state, collecting measurements, classifying examples, or fixing a broken process.

Give enough setup to start, but don't give the final answer.

Time

A 10-minute activity should have one main goal. A 20-30 minute activity may include setup, execution, and reflection.

Computing Teaching Reference Handbook

Programming foundations

Variables hold values that can change while a program runs. A beginner lesson should connect the variable name to its purpose, not just show assignment syntax. Useful examples include a score, temperature, username, or running total.

Conditions choose between paths. Learners often confuse assignment with comparison, or forget that only one branch may run in a simple if/elif/else chain. Short trace tables help make the control flow visible.

Loops repeat work. A for-loop is useful when iterating over a known collection or range. A while-loop is useful when repetition depends on a condition. A common beginner error is creating a while-loop whose condition never changes.

Functions group reusable logic. Learners should distinguish defining a function from calling it, parameters from arguments, and print from return. Examples should keep side effects small so the return value is easy to see.

Lists hold ordered values. Beginner work should cover indexing, iteration, adding/removing items, and the fact that index 0 refers to the first item.

Git and collaboration

A Git repository stores the project history. A branch gives a developer a place to work without changing the main branch immediately. Commits should represent understandable changes rather than one huge dump at the end of a week.

A merge conflict is not a broken repository. It means Git found overlapping edits and needs a human to decide the final content. Learners should read both sides of the conflict before editing the file.

A pull request is a review point. A useful PR description says what changed, how to test it, and anything still unfinished. Secrets such as API keys do not belong in commits.

Adding a file to .gitignore prevents future tracking when configured correctly, but it does not erase a secret that was already committed. Exposed keys should be rotated.

SQL basics

SELECT chooses columns or expressions to return. WHERE filters rows before grouping. JOIN combines rows from related tables. GROUP BY creates groups for aggregate functions such as COUNT, SUM, AVG, MIN, and MAX.

A query can be syntactically valid and still be logically wrong. Joining users.id to orders.id instead of orders.user_id may return believable data by accident. Examples should include this kind of mistake.

SQLite is useful for small local applications because it has no separate server process. It still uses relational tables and SQL, so the same core query ideas carry over to larger databases.

REST and FastAPI

An HTTP request has a method, URL, headers, and sometimes a body. A response has a status code, headers, and usually a body. The method should match the action being performed.

FastAPI maps routes to Python functions. Pydantic models describe the expected input shape. When a required field is missing, validation should fail before business logic runs.

Status codes are part of the API contract. A missing resource is different from a server crash, so 404 and 500 communicate different problems.

Practice ideas

Ask learners to trace a short program before they write one. Give a broken SQL JOIN and ask what data it would connect. Give two API requests and ask which one should be GET versus POST. Give a small Git history and ask where a secret leak occurred.

Good computing lessons should make internal state visible: variable values, loop iterations, request bodies, response JSON, SQL rows, and Git history.

Mathematics Teaching Reference Handbook

Linear equations

An equation states that two expressions are equal. Solving means finding values that keep the equality true. When the same operation is applied to both sides, equality is preserved.

One-step equations isolate the variable with one inverse operation. Two-step equations require learners to decide the order of undoing operations. Checking by substitution should be included so the process does not become mechanical.

Common mistakes include changing only one side, losing a negative sign, and applying an operation to one term instead of an entire side.

Graphs and slope

A coordinate graph connects numeric relationships to position. Slope measures vertical change divided by horizontal change. Learners should calculate slope from two points and interpret what the sign means.

A positive slope rises from left to right, a negative slope falls, a zero slope is horizontal, and a vertical line has undefined slope. The y-intercept is where x is zero.

Graph questions should include interpretation, not only calculation. For example, a slope of 3 in a savings graph may mean an increase of three units per time step.

Quadratics

A quadratic expression contains a squared variable term. Learners may expand simple binomials, factor common patterns, and connect factors to zeros.

The discriminant can be introduced as a way to anticipate the number of real roots without requiring a full proof. The emphasis should remain on what the value tells us.

Graph shape and algebra should be linked carefully. A parabola opening upward does not by itself tell the learner whether there are zero, one, or two x-intercepts.

Data and statistics

Mean uses every value and is sensitive to outliers. Median is the middle ordered value and may better represent a skewed dataset. Mode is the most frequent value and may be absent or have multiple values.

Charts should be read critically. A truncated axis can exaggerate a small difference. Percent change and percentage points are not the same thing.

Activities should use small datasets that learners can check by hand before moving to larger tables.

Science Teaching Reference Handbook

Physics

Distance is total path length while displacement includes direction from start to finish. Speed is distance per time, while velocity includes direction. Acceleration describes change in velocity over time.

Newton's first law describes persistence of motion when net force is zero. The second law connects net force, mass, and acceleration. The third law describes equal and opposite interaction forces acting on different objects.

Work involves force acting through a displacement. Kinetic energy depends on motion, while gravitational potential energy depends on position in a gravitational field.

Physics lessons should include units and interpretation. A correct numeric answer without units is incomplete in many contexts.

Biology

Cells contain structures with specialized functions. The membrane controls movement of substances, the nucleus contains genetic material in eukaryotic cells, and mitochondria are involved in cellular respiration.

DNA contains genetic information. During transcription, a DNA template is used to make RNA. During translation, ribosomes read mRNA codons to build a protein.

Photosynthesis stores energy in chemical form using light, carbon dioxide, and water. Cellular respiration releases usable energy from chemical fuels. The processes are related but should not be described as simple opposites in every detail.

Chemistry

Atomic number is the number of protons. Isotopes of the same element have the same proton count but different neutron counts. Ions form when the electron count changes relative to the proton count.

Chemical equations represent rearrangements of atoms. Balancing changes coefficients, not the chemical formulas of the substances.

Acids and bases can be introduced through pH and observable behavior. Safety matters when discussing real substances, and classroom examples should not encourage unsupervised experiments.

Science activities

Ask learners to predict, observe, and explain. A short table of measurements can be more useful than a long lecture. Activities should distinguish evidence from explanation.

Diagrams and formulas should be paired with plain-language interpretation so learners understand what the symbols represent.

Business Communication Reference Handbook

Professional email

A useful subject line tells the reader what the message is about. The first paragraph should make the purpose clear rather than spending several sentences on background.

Requests should be specific. 'Please send the revised file by Thursday afternoon' is easier to act on than 'please update me soon'.

Tone should match the relationship and situation. Direct language can still be polite. Long greetings and excessive formality often make routine work emails harder to scan.

Project updates

A short update usually needs context, completed work, current work, blockers, and next steps. The exact order can change, but the reader should quickly see whether anything needs attention.

A blocker should name what is blocked and what is needed. 'Waiting on API access from the client' is more useful than 'blocked'.

Don't turn an update into a diary of every action taken. Include details that change decisions or next steps.

Meeting notes

Useful notes capture decisions, owners, deadlines, and open questions. A transcript records everything said, but notes should select what matters after the meeting.

Action items should be written so another person can understand the task without replaying the meeting.

Where a decision is tentative, the notes should say so instead of presenting it as final.

Communication Reference Notes

Professional writing

A useful professional email makes the purpose clear early, gives enough context, and ends with a specific request or next step.

Project updates should separate completed work, current work, blockers, and next steps so a reader can scan them quickly.

Meetings

Meeting notes should capture decisions, owners, dates, and unresolved questions. A transcript isn't the same as useful meeting notes.

Computing Reference Notes

Python basics

Variables store values, conditions choose paths, loops repeat work, functions group reusable logic, and lists hold ordered collections.

In Git, a branch isolates work. Commits record logical changes. Merge conflicts happen when Git can't combine overlapping edits automatically.

SQL and APIs

A SQL JOIN combines related rows using a condition. GROUP BY collects rows so aggregates such as COUNT or SUM can be calculated.

REST APIs expose routes that accept HTTP requests. Request validation should happen before business logic where practical.

Mathematics Reference Notes

Equations and graphs

Solving an equation means finding a value that makes both sides equal. Operations used to isolate a variable should preserve equality.

Slope describes change in y for a change in x . The y -intercept is where a line crosses the vertical axis.

Quadratics and data

Quadratic expressions include a squared term. Factoring can reveal zeros when a quadratic can be written as a product.

Mean is sensitive to outliers, while median is often more stable when extreme values are present.

Science Reference Notes

Physics

Speed describes how fast distance changes. Velocity includes direction. Acceleration describes how velocity changes over time.

Net force determines acceleration according to the relationship between force and mass. Energy can move between kinetic, potential, thermal, and other forms.

Biology and chemistry

Cells contain structures with different functions. DNA stores genetic information that can be transcribed into RNA and translated into protein.

Atoms contain protons, neutrons, and electrons. Chemical reactions rearrange atoms while conserving the number of atoms of each element.